

4.10.12

EE25BTECH11011 - Navya Banavathu

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Question

The straight line $5x + 4y = 0$ passes through the point of intersection of the straight lines $x + 2y - 10 = 0$ and $2x + y + 5 = 0$.

Solution

The equations of the given lines are

$$\begin{pmatrix} 1 & 2 \end{pmatrix} \mathbf{x} = 10 \quad (1)$$

$$\begin{pmatrix} 2 & 1 \end{pmatrix} \mathbf{x} = -5 \quad (2)$$

$$\begin{pmatrix} 5 & 4 \end{pmatrix} \mathbf{x} = 0 \quad (3)$$

Forming a single matrix equation, we have

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 5 & 4 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 10 \\ -5 \\ 0 \end{pmatrix} \quad (4)$$

The augmented matrix is

$$\left(\begin{array}{cc|c} 1 & 2 & 10 \\ 2 & 1 & -5 \\ 5 & 4 & 0 \end{array} \right) \quad (5)$$

Solution

Applying row operations:

$$\left(\begin{array}{cc|c} 1 & 2 & 10 \\ 2 & 1 & -5 \\ 5 & 4 & 0 \end{array} \right) \xrightarrow{R_2-2R_1, R_3-5R_1} \left(\begin{array}{cc|c} 1 & 2 & 10 \\ 0 & -3 & -25 \\ 0 & -6 & -50 \end{array} \right) \quad (6)$$

$$\xrightarrow{R_2/(-3)} \left(\begin{array}{cc|c} 1 & 2 & 10 \\ 0 & 1 & 25/3 \\ 0 & -6 & -50 \end{array} \right) \quad (7)$$

$$\xrightarrow{R_1-2R_2, R_3+6R_2} \left(\begin{array}{cc|c} 1 & 0 & -20/3 \\ 0 & 1 & 25/3 \\ 0 & 0 & 0 \end{array} \right) \quad (8)$$

Solution

Thus, the RREF gives the solution vector

$$\mathbf{x} = \begin{pmatrix} -\frac{20}{3} \\ \frac{25}{3} \end{pmatrix} \quad (9)$$

Hence, The straight line $5x + 4y = 0$ passes through the point of intersection of the straight lines $x + 2y - 10 = 0$ and $2x + y + 5 = 0$.

Code (C)

Part 1

```
#include <stdio.h>

int find_intersection(double *x, double *y) {
    double a1 = 1, b1 = 2, c1 = 10;
    double a2 = 2, b2 = 1, c2 = -5;

    double det = a1*b2 - a2*b1;
    if(det == 0) return 0; // parallel

    *x = (c1*b2 - c2*b1) / det;
    *y = (a1*c2 - a2*c1) / det;

    if(5*(*x) + 4*(*y) == 0)
        return 1; // passes
    else
        return 2; // does not pass
}
```

Python Code: Import and Define points

Part 1

```
import ctypes
import matplotlib.pyplot as plt
import numpy as np

# --- Load the DLL ---
lib = ctypes.CDLL(r"C:\Users\Navya\code8.dll")

# Prepare variables to receive x and y from C
x = ctypes.c_double()
y = ctypes.c_double()

# Call the C function
result = lib.find_intersection(ctypes.byref(x), ctypes.byref(y))
xi, yi = x.value, y.value
```

Part 2

```
print(f"Intersection point: ({xi:.2f}, {yi:.2f})")

if result == 0:
    print("Lines are parallel, no intersection.")
elif result == 1:
    print("The line  $5x + 4y = 0$  passes through the intersection point.")
else:
    print("The line  $5x + 4y = 0$  does NOT pass through the intersection point.")
```


Python Code: Plotting

Part 3

```
# Plot the lines
```

```
x_vals = np.linspace(-10, 10, 400)
```

```
# Line 1:  $x + 2y = 10 \Rightarrow y = (10 - x)/2$ 
```

```
y1 = (10 - x_vals)/2
```

```
# Line 2:  $2x + y = -5 \Rightarrow y = -5 - 2x$ 
```

```
y2 = -5 - 2*x_vals
```

```
# Line 3:  $5x + 4y = 0 \Rightarrow y = -5/4 * x$ 
```

```
y3 = -5/4 * x_vals
```

```
plt.figure(figsize=(8,6))
```

```
plt.plot(x_vals, y1, label=r'$(1\ 2)\mathbf{x} = 10$', color='blue',  
        )
```

```
plt.plot(x_vals, y2, label=r'$(2\ 1)\mathbf{x} = -5$', color='green',  
        )
```

```
plt.plot(x_vals, y3, label=r'$(5\ 4)\mathbf{x} = 0$', color='red')
```

Part 4

```
# Mark the intersection point
plt.scatter(xi, yi, color='black', zorder=5)
plt.text(xi+0.5, yi, f'({xi:.2f},{yi:.2f})', fontsize=10)

plt.xlabel('x')
plt.ylabel('y')
plt.title('Intersection of Lines')
plt.grid(True)
plt.legend()
plt.xlim(-10, 10)
plt.ylim(-10, 10)

# Save figure as fig8.png
plt.savefig('fig8.png', dpi=300)
plt.show()
```

Graphical Representation

